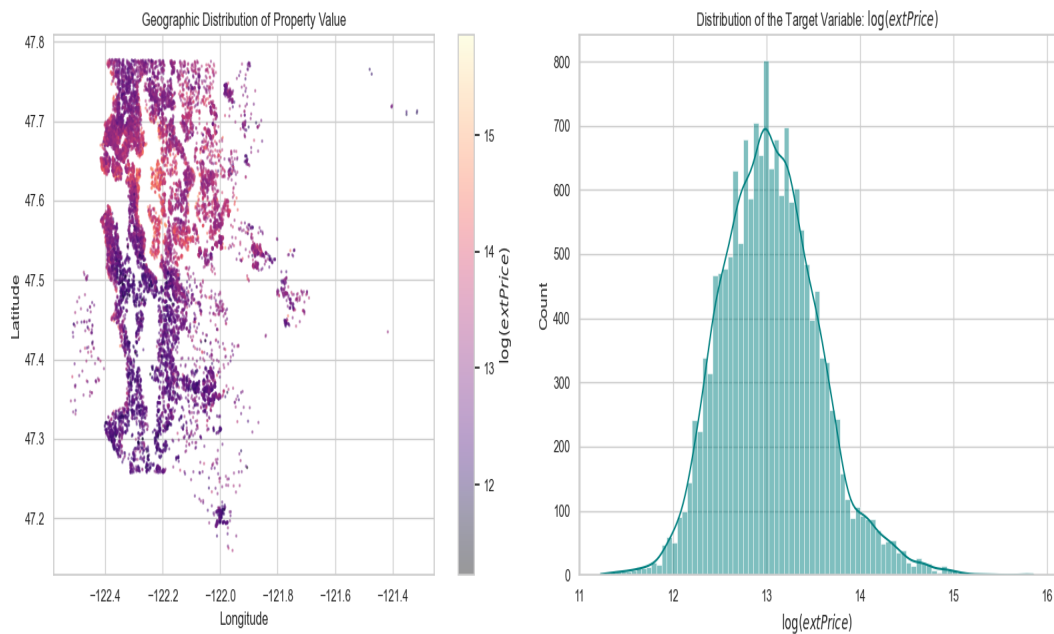


Satellite Imagery-Based Property Valuation

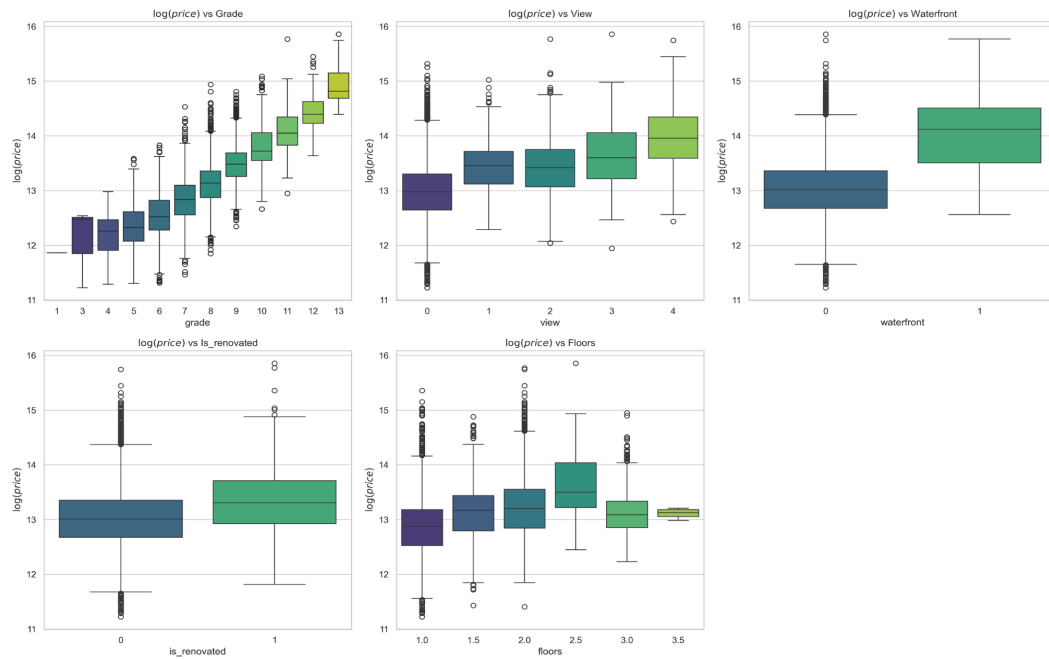
Overview

This project builds a multimodal regression system to predict house prices using both tabular housing attributes and spatial context derived from map-based imagery. The goal is to enhance traditional valuation models by incorporating neighborhood and environmental signals.

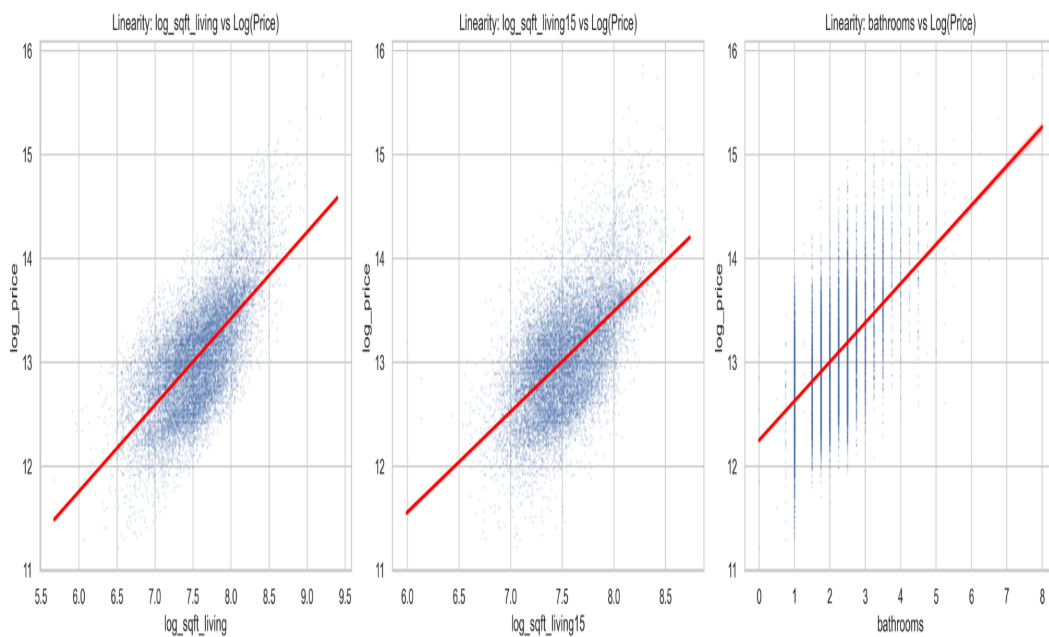
Exploratory Data Analysis



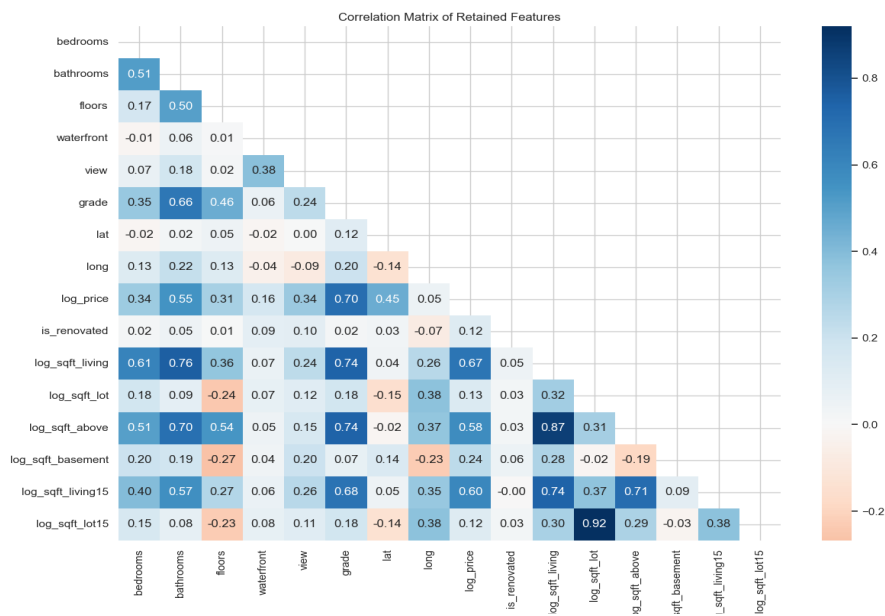
Geographic distribution of property values and target distribution.



Effect of categorical variables on $\log(\text{price})$.



Linearity analysis of continuous features.



Correlation matrix of retained features.

Modeling & Results

A baseline tabular regression model was compared against a multimodal model that incorporates spatial visual context. The multimodal approach showed improved RMSE and R^2 , demonstrating the value of spatial information.

Conclusion

The results confirm that integrating spatial context improves property valuation accuracy. This approach is scalable and can be extended to other regions and higher-resolution imagery.